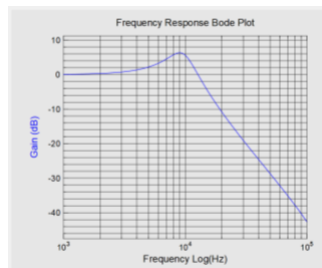


HW Subsystem Test Plan

Sensor board is not connected to the Microcontroller-board, References: HW_Testing.docx (PM3), Checklist_Marks.xlsx (PM4)

Test item, Description	Test Conditions	Expected, Tolerance	Measured Result	Comparison, Verdict	Remarks, Discussion
Signal Gain-Phase - w/o radar sensor - on I- and Q-paths	Sine input signal: 50 uVpp & 1mVpp 10 Hz ...	Gain in passband: ... dB $\pm$... dB cut-off frequencies LP/HP: ...	Scope Option: Bode Plots (see example)	passed / as designed not passed	...
DC level - w/o radar sensor - on I- and Q-paths - after each Amp					
Radarsignal Gain-Phase - on I- and Q-paths	Radar Doppler Target K-DT1 with $f_d = \dots$ at distance ...	I/Q-Gain-Difference ... I/Q-Phase difference $90^\circ \pm \dots$	Scope Channel A/B		
Additional Features					

Example: Frequency Response Bode Plot (PicoScope 2206B: <https://www.picotech.com/downloads>,
 3rd party WINDOWS app: FRA For PicoScope: <https://bitbucket.org/hexamer/fra4pico/branch/master>)



FW Subsystem Test Plan

Microcontroller-board is not connected to the Sensor board, References: HW_Testing.docx (PM3), Checklist_Marks.xlsx (PM4)

Test item, Description	Test Conditions	Expected, Tolerance	Measured Result	Comparison, Verdict	Remarks, Discussion
ADC Samples - on I- and Q-paths	no signal ...	DC-value of virtual GND ...		passed / as designed not passed	...
FFT output - on display	Sine input signal: ... Hz, ... Vpp	peak and leakage			
Frequency / Speed - w/o sign! - on I- and/or Q-paths	Sine input signal: ... Hz, ... Vpp	Frequency $+... \pm \Delta f/2$ ($\Delta f = f_s/NFFT$)	speed or frequency without sign		
Frequency / Speed - with sign	sin/cos signals from Matlab and USB-Audio Interface: ... Hz, ... Vpp	Frequency $... \pm \Delta f/2$ ($\Delta f = f_s/NFFT$)	speed or frequency with sign		
Additional Features					